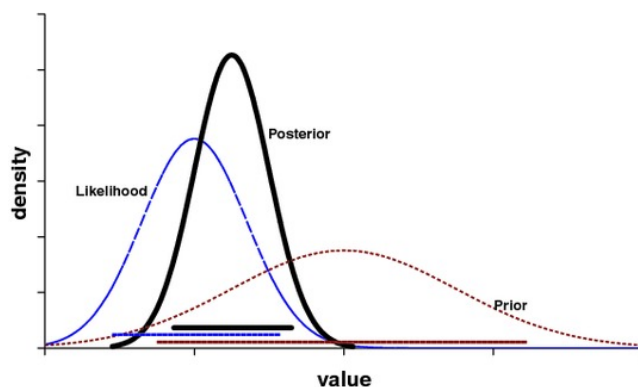


## Tutorial Problems: Week 2

These exercises give you some practice in deriving posterior distributions, and evaluating quantities of interest from them.

### 1. *Posterior practice*



- (a) Derive the posterior distribution if  $x_1, \dots, x_n$  are a random sample from the distribution with probability mass function

$$p(x|\theta) = \theta^{x-1}(1 - \theta); \quad x = 1, 2, \dots$$

and assuming a  $\text{Beta}(p, q)$  prior distribution

$$\pi(\theta) = \frac{\theta^{p-1}(1 - \theta)^{q-1}}{\text{Be}(p, q)}, \quad 0 < \theta < 1,$$

where  $\text{Be}(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p + q)$  and  $\Gamma$  is the gamma function.

- (b) Derive the posterior distribution if  $x_1, \dots, x_n$  are a random sample from the distribution with probability mass function

$$p(x|\theta) = \frac{e^{-\theta}\theta^x}{x!}, \quad x = 0, 1, \dots$$

and assuming a prior distribution

$$p(\theta) = e^{-\theta}, \quad 0 < \theta.$$

## 2. Defective items



- (a) The proportion,  $0 \leq \theta \leq 1$ , of defective items in a large shipment is unknown, but expert assessment assigns  $\theta$  the Beta(2,200) prior distribution. If 100 items are selected at random from the shipment, and 3 are found to be defective, what is the posterior distribution of  $\theta$ ?
- (b) If a different analyst, having observed the 3 defectives, calculated her posterior distribution as being a Beta distribution with mean  $4/102$  and variance  $0.0003658$ , then what prior distribution had she used?

## 3. Component variability



The diameter of a component from a long production run varies according to a  $N(\theta, 1)$  distribution. An Engineer specifies that the prior distribution for  $\theta$  is  $N(10, 0.25)$ . In one production run 12 components are sampled and found to have a sample mean diameter of  $31/3$ . Use this information to calculate the probability that the mean component diameter is at least 10 units.

## 4. Magnetic tape



The number of defects in a 1200 metre roll of magnetic tape has a  $\text{Poisson}(\theta)$  distribution. Suppose that the prior distribution for  $\theta$  is  $\text{Gamma}(3,1)$ . When 5 rolls of this tape are selected at random, the number of defects found on each are 2, 2, 6, 0 and 3 respectively. Determine the posterior distribution of  $\theta$ .

5. Some well known prior-likelihood conjugate pairs are:

| Likelihood                                                  | Prior                | Posterior                                                     |
|-------------------------------------------------------------|----------------------|---------------------------------------------------------------|
| $x \sim \text{Bin}(n, \theta)$                              | $\text{Beta}(p, q)$  | $\text{Beta}(p + x, q + n - x)$                               |
| $x_1, \dots, x_n \sim \text{Poisson}(\theta)$               | $\text{Gamma}(p, q)$ | $\text{Gamma}(p + \sum_{i=1}^n x_i, q + n)$                   |
| $x_1, \dots, x_n \sim N(\theta, \tau^{-1})$ ( $\tau$ known) | $N(b, c^{-1})$       | $N(\frac{cb + n\tau\bar{x}}{c + n\tau}, \frac{1}{c + n\tau})$ |
| $x_1, \dots, x_n \sim \text{Gamma}(k, \theta)$ ( $k$ known) | $\text{Gamma}(p, q)$ | $\text{Gamma}(p + nk, q + \sum_{i=1}^n x_i)$                  |
| $x_1, \dots, x_n \sim \text{Geometric}(\theta)$             | $\text{Beta}(p, q)$  | $\text{Beta}(p + n, q + \sum_{i=1}^n x_i - n)$                |
| $x \sim \text{NegBin}(r, \theta)$                           | $\text{Beta}(p, q)$  | $\text{Beta}(p + r, q + x)$                                   |

Derivations for the first four have been provided in the Week 2 Lecture notes. Show that the remaining two conjugate relationships hold. Specifically, for:

- The geometric distribution  $\text{Geom}(\theta)$ , which has a pmf proportional to  $(1 - \theta)^{x-1}\theta$ , where  $0 \leq \theta \leq 1$  is the probability of success, and  $x = 1, 2, \dots$  is the number of failures until the first success.
- The Negative Binomial distribution  $\text{NegBin}(r, \theta)$ , which has a pmf proportional to  $\theta^r(1 - \theta)^x$  where  $0 \leq \theta \leq 1$  is the success probability,  $r$  = number of successes required until the experiment is stopped and  $x = 0, 1, \dots$  number of failures until  $r > 0$  successes occur. ( $x$  is the variable,  $r$  is a parameter).

6. Find Jeffreys' prior for  $\theta$  in the geometric model

$$p(x|\theta) = (1 - \theta)^{x-1}\theta; \quad x = 1, 2, \dots$$

for a random sample  $x_1, \dots, x_n$ .

7. Suppose you toss a coin  $n$  times and observe  $x$  heads. Your goal is to estimate  $\theta$ , the probability of throwing a head. Compute the maximum likelihood estimate and perform a Bayesian analysis with uniform prior on  $\theta$ . Compare the values of the MAP, MLE and posterior mean.